

Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

- A. Decreased prolactin, decreased FSH, decreased LH
- B. Decreased prolactin, increased FSH, increased LH
- C. Increased prolactin, decreased FSH, increased LH
- D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

<think>
Finding Reasoning Paths :
1 . The presence of hyperprolactinemia is a key clue , as it can cause amenorrhea and galactorrhea (milky nipple discharge).
2 . Galactorrhea suggests stimulation of the mammary glands , which could be mediated by hyperprolactinemia or a pituitary adenoma .
3 . Amenorrhea and galactorrhea could also be due to a hypothalamic cause (e.g ., hormone imbalance or pituitary pathology).
4 . Hyperprolactinemia can result from various causes , including pituitary adenomas , hypothyroidism , or medications .
5 . The patient's symptoms of amenorrhea and galactorrhea are characteristic of a hyperprolactinemic state , which typically leads to decreased FSH and LH levels due to direct or indirect suppression of the hypothalamic-pituitary-gonadal axis .

Reasoning Process :
1 . **Symptom Analysis** :
- The patient presents with amenorrhea and galactorrhea , which are classic symptoms of hyperprolactinemia .
- Prolactin inhibits the release of gonadotropin-releasing hormone (GnRH) from the hypothalamus , leading to decreased secretion of FSH and LH from the pituitary gland .

2 . **Hyperprolactinemia Causes** :
- The most common cause of hyperprolactinemia in a reproductive-aged woman is a prolactin-secreting pituitary adenoma (prolactinoma).
- Hypothyroidism , medications (e.g ., certain psychiatric drugs , phenothiazines , and neuroleptics), and hypothalamic dysfunction can also cause hyperprolactinemia .

3 . **Gonadotropin Response** :
- In hyperprolactinemia , the pituitary gland is inhibited from releasing FSH and LH , leading to decreased levels of these hormones .
- Gonadotropin levels are directly regulated by prolactin , and its elevation suppresses their production .

4 . **Exclusion of Other Causes** :
- While hypothyroidism and other hypothalamic conditions can cause amenorrhea and galactorrhea , they do not directly cause hyperprolactinemia .
- The patient's symptoms are most consistent with hyperprolactinemia , which typically manifests with decreased FSH and LH levels .

5 . **Pathophysiology of Hyperprolactinemia** :
- The increased prolactin levels suppress pituitary gonadotropin (FSH and LH) production , leading to the observed laboratory findings .

Conclusion :
The laboratory findings in this patient are expected to show :
- **Decreased FSH** (due to direct suppression by hyperprolactinemia)
- **Decreased LH** (due to direct suppression by hyperprolactinemia)

This conclusion aligns with the pathophysiology of hyperprolactinemia and its effect on the hypothalamic-pituitary-gonadal axis .
</think>
<answer>
Decreased prolactin , decreased FSH , decreased LH
</answer>